Analytics & Big Data

Session 4: Regression I

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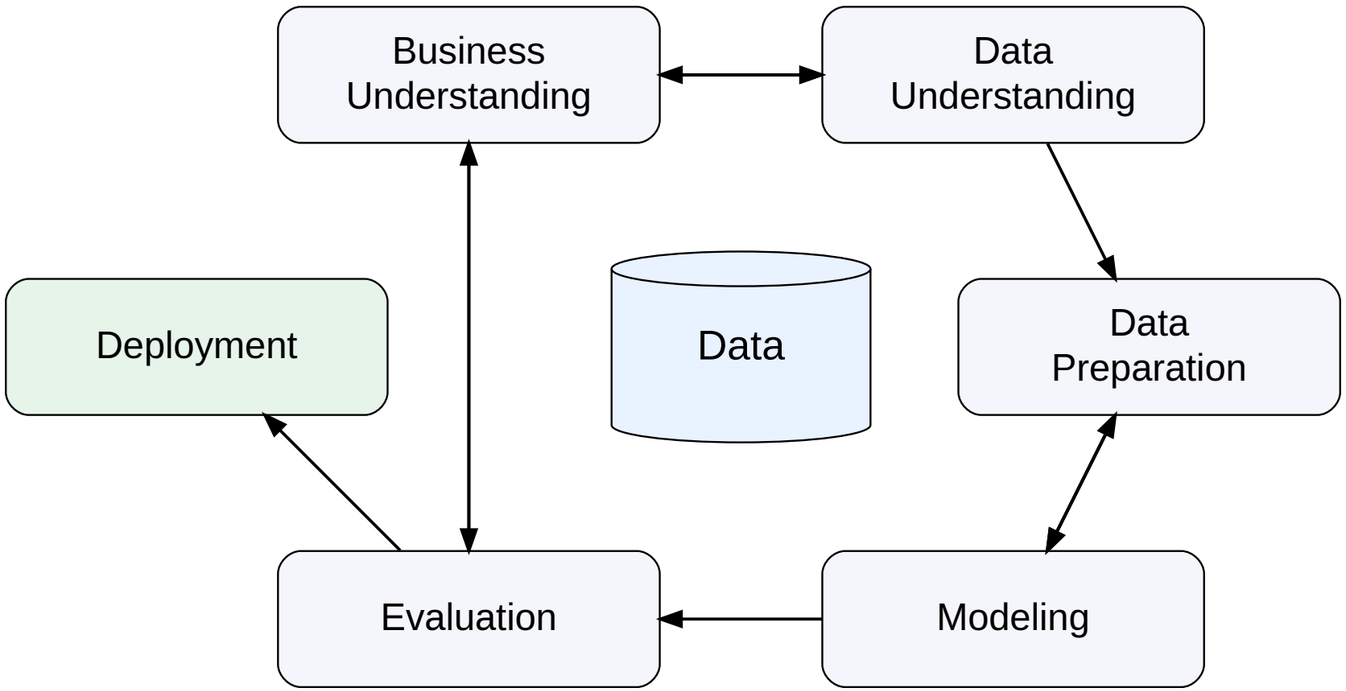
Learning objectives

- **Explain the stages of a model-based analytics workflow** using linear regression as an example.
- **Understand and interpret linear regression models**, including how they represent relationships between variables, how coefficients are interpreted, how OLS estimates parameters, and how model quality is assessed (e.g., via R^2 and overfitting considerations).
- **Describe how regression models are implemented in Python**, mapping conceptual components of the analytics process to their realization in scikit-learn.

Process


Business Understanding
Data Understanding
Deployment
Data
Data Preparation
Evaluation
Modeling

Business understanding

Case: House Prices — What Drives Property Value?

Housing markets represent one of the largest asset classes in most economies. Residential real estate accounts for a substantial share of household wealth, and even small pricing errors can translate into large financial consequences.

Understanding what drives house prices is therefore highly relevant:

- **Real estate firms** use pricing models to advise clients.
- **Banks** rely on valuation models to assess collateral and manage mortgage risk.
- **Insurers and policymakers** use property data for risk assessment, taxation, and urban planning.

This case illustrates how regression analysis transforms intuition about property value into systematic, evidence-based decision support across the housing ecosystem.

Data understanding

Dataset: Ames Housing (Kaggle)

Dataset: Ames Housing Dataset

Kaggle is a widely-used online platform for data science and machine learning. It provides:

- Public datasets
- Notebooks for analysis
- Competitions and community discussions

Task (5–7 minutes)

05:30

1. Navigate to the dataset page on Kaggle.
2. Skim the dataset description.
3. Inspect the list of variables (features).
4. Open the data preview and:
    - Identify the target variable.
    - Find at least three potential explanatory variables.

Be prepared to briefly discuss:

- Which variables you expect to be strong predictors of price.

Data preparation

…

We skip data preparation for now…

Modeling

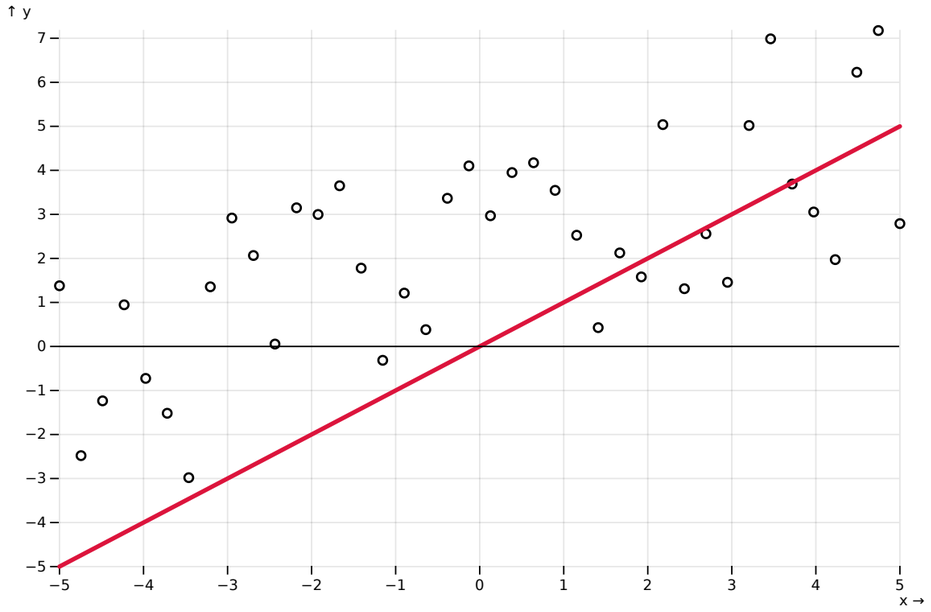
Regression models: a visual illustration

β_0 and β_1 move the line

β_0 (intercept) 0

β_1 (slope) 1

Plot Data



Model (OLS)

illustrate the optimization process

Model results

interpret coefficients….

Evaluation

R^2 (does it help to predict?)

TBD: illustrate overfitting?!

TBD: other considerations? (generalization/robustness, ethics, …)

Deployment

TODO / TBD.

- Include a prediction example (based on a formula, python code).
- Also refer to the describe/predict/prescribe
- TBD: how can it be used to prescribe?

Vocabulary

A **method** is a composition of formalized principles that form the basis for a stringent calculation process.

An **algorithm** is a procedure or set of steps or rules to accomplish a task. It is usually the implementation of a method. Algorithms are used to build models.

In the given context, a **model** is the description of the relationship between variables. It is used to create output data from given input data, for example to make predictions.

Fitting a model means that you estimate the model using the observed data. You are using your data as evidence to help approximate the real-world mathematical process that generated the data. Fitting the model often involves optimization methods and algorithms, such as maximum likelihood estimation, to help get the parameters.

Overfitting is the term used to mean that you used a dataset to estimate the parameters of your model, but your model isn't that good at capturing reality beyond your sampled data.

— Source: Schutt/O'Neil (2013): Doing Data Science.

Summary

- Regression is a core example of a structured, model-based analytics process: from business question to data, modeling, evaluation, and potential deployment.
- Linear regression models quantify relationships between variables through coefficients, estimated via optimization (OLS) and evaluated using metrics such as R^2 while considering generalization and overfitting.
- In practice, the analytics workflow can be implemented in Python: define a model, fit it to data, generate predictions, and evaluate performance — a pattern that extends to many other modeling approaches.

Note: more on python in the exercises